

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9747/03

Paper 3

18 September 2009

Applications

1 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

This document consists of **10** printed pages.



- 1 In flax plants, the flowers are white, lilac or blue. Fig.1 shows the pathway by which the flower cells produce coloured pigments.

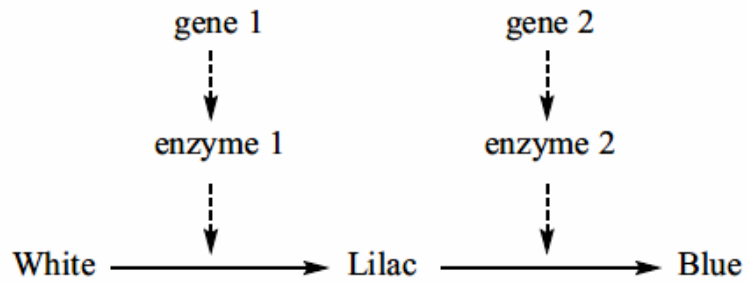


Fig. 1

- (a)(i) A single base deletion occurs in gene 1. Describe and explain the effect of this mutation on the colour of flax plant flowers.

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[4]

- (b) Gel electrophoresis was used to separate the enzymes involved in this pathway. When extracts of the differently coloured flax petals were analysed, four different patterns of bands were produced. In the table, only bands that contain functional enzymes are shown.

Result of electrophoresis	Colour of petal
	White
	
	
	

- (i) With reference to Fig. 1, complete the table to give the colour of the petal from which each extract was taken. [3]
- (ii) Describe how the enzymes were separated by gel electrophoresis.

[5]

- (c) Genomic libraries and cDNA libraries can be created for the flax plant genome. State **three** differences between the two gene libraries.

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..... [3]

[Total: 15]

- 2 Children with severe combined immunodeficiency disorder (SCID) cannot produce the many types of white blood cells that fight infections. This is because they do not have the functional gene to make the enzyme ADA.

Some children with SCID have been treated with stem cells. Stem cells can divide and develop into any type of blood cell, including white blood cells. The treatment used with the children is described in Fig. 2.1.

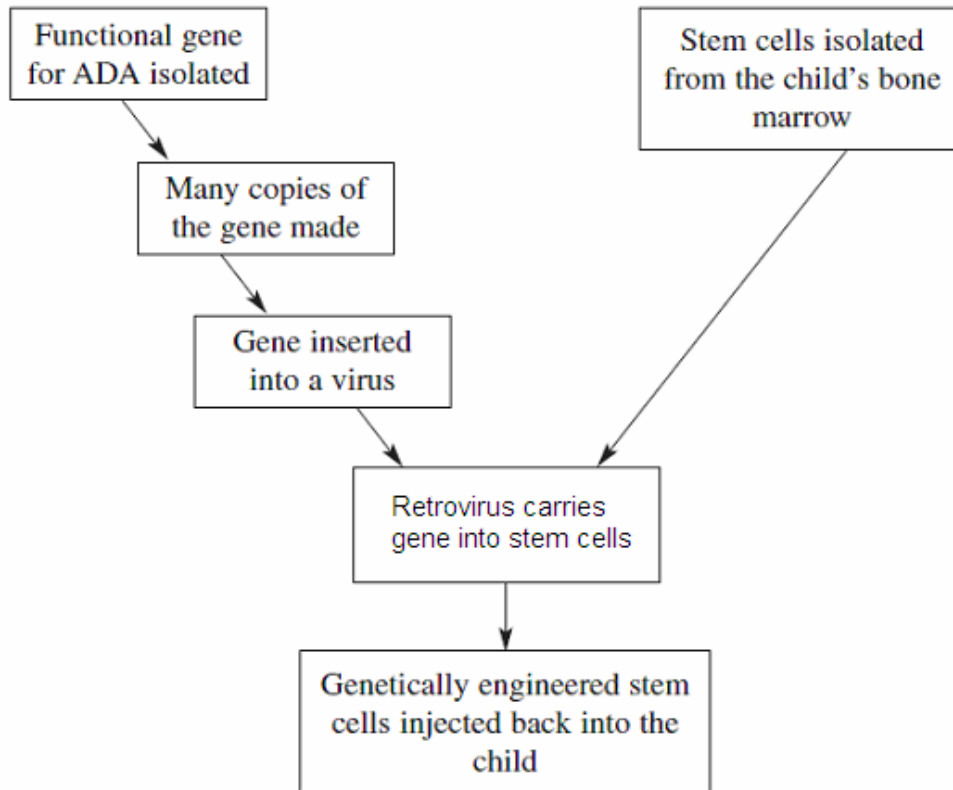


Fig. 2.1

- (a)(i) Explain how the lack of a functional gene coding for ADA leads to SCID.

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..... [2]

- (ii) Suggest and explain **two** reasons why stem cells were used in this treatment.

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..... [2]

- (iii) Suggest what stem cells are isolated from the bone marrow.

..... [1]

- (iv) Describe how many copies of functional ADA gene may be produced in the process shown in Fig.2.1.

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..... [3]

- (v) State **two** advantages of using virus to carry the gene into the stem cells.

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..... [2]

- (vi) Suggest why the method described in Fig. 2.1 is better than transplanting stem cells with functional ADA gene from a donor.

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..... [1]

- (b) Child A and Child B were treated with genetically engineered stem cells described in Fig. 2.1. Fig. 2.2 shows the number of functioning white blood cells in both children during the year following treatment. Children who do not suffer from SCID have between 5000 and 8000 white blood cells per mm^3 of blood.

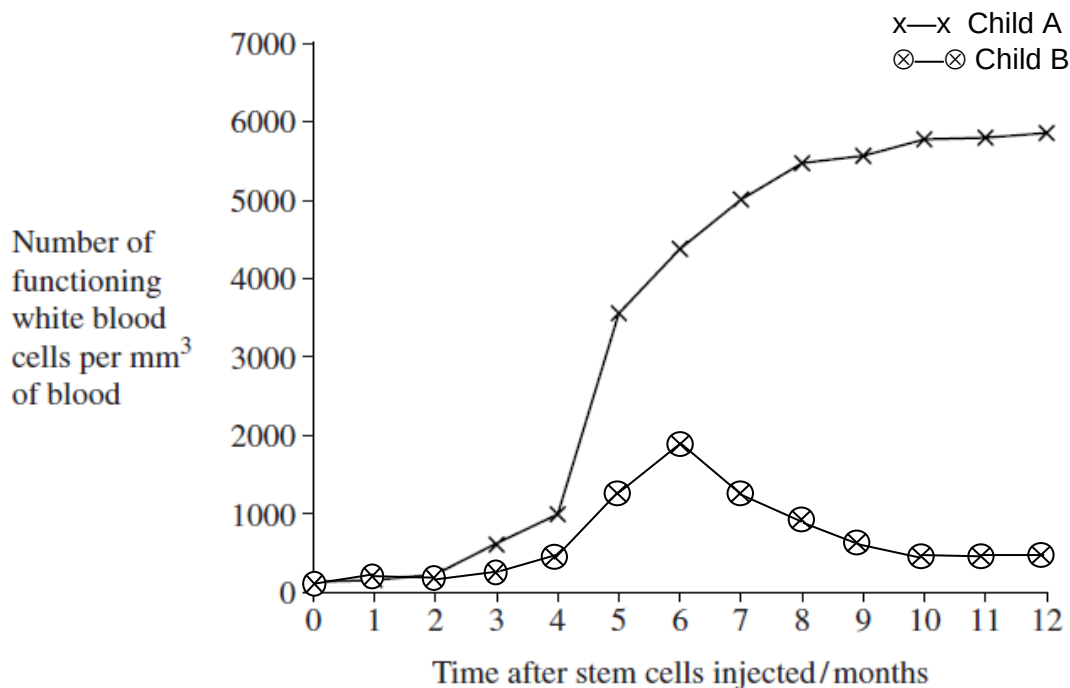


Fig. 2.2

- (i) Describe the results of the treatment.

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[2]

- (ii) Explain the difference in response between the two children to treatment.

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[2]

[Total: 15]

- 3** Genetic engineering is a technique that has been developed to improve the quality and yield of many crop plants and animals.

Numbers of genetically modified (GM) plants can be increased by cloning from tissue culture.

- (a)** What do you understand by '*genetically modified (GM) plants*'?

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..... [1]

- (b)** Describe the process of cloning such plants.

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..... [3]

Maize can be genetically modified to produce a toxin, which acts as an insecticide, killing any insects which feed on the plants.

Growth of these genetically modified maize plants suggested that the toxin might also kill a range of beneficial soil organisms, such as insects, worms, fungi and bacteria.

To investigate this, seeds of the maize were surface sterilized and germinated on sterile agar. The seedlings were then grown at a constant temperature in both sterile and non-sterile soil-free nutrient medium in closed containers.

After 6, 12 and 18 days the plants were transferred to fresh medium. The concentration of the toxin in the medium was measured. The larvae of several insect species were put in contact with it. The results were summarized in Table 3.1.

Table 3.1

time/days	6	12	18	6	12	18
medium	sterile	sterile	sterile	not sterile	not sterile	not sterile
toxin in medium/ a.u.	8	10	12	1	0	0
% mortality of insects after 5 days of exposure to medium	90	95	99	0	0	0

(c) With reference to Table 3.1,

(i) describe a suitable control for this experiment

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..... [1]

(ii) describe and explain the results under sterile conditions.

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(iii) discuss the extent to which the results confirm that growing the genetically modified maize will damage the beneficial soil organisms.

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- (d) Discuss briefly the ethical and social implications of genetically modified organism.

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..... [4]

[Total: 15]

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

Question 4

- (a) Describe the functions of **three** enzymes needed in the formation of recombinant DNA. [6]
- (b) Describe how nucleic acid hybridization can be used in RFLP analysis on isolated DNA sample. [6]
- (c) Explain the application of RFLP analysis in disease detection and DNA fingerprinting. [8]

[Total: 20]